

Semester Project Description

NE 511

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Abstract

This report introduces the Nonlinearly Implicit Low-Order CMFD (NILO-CMFD) method and the Multilevel Nonlinear Projective (MNP) method for solving multi-physics problems involving coupled neutron transport and Thermal Hydraulics. Both methods could be applied to any TH solver, but the MNP method, like X-CMFD, requires repeated inversion of the TH operator, which may potentially very computationally expensive. For my NE511 computational project I propose a NILO-MNP method which combines the advantages of these existing methods.

For the project, I will at minimum implement the MNP method, and if time allows I will implement the NILO-MNP method, and perform Fourier stability analysis on both methods.

1 Introduction

The NILO-CMFD method uses nonlinear diffusion acceleration (NDA) with coarse-mesh finite difference (CMFD) to accelerate coupled transport and TH. It also uses an approximate TH operator within inner iterations to reduce computational cost without introducing approximation in the final result.

The MNP method uses Multigroup and Grey Quasidiffusion to accelerate the same class of problems. In single-physics applications, Multilevel Quasidiffusion converges faster than NDA, but Fourier analysis has not been performed for the multiphysics case of MLQD.

This report summarizes the NILO-CMFD and MNP methods for coupled transport and TH. For my NE511 computational project I plan to implement the MNP method, first in the steady-state case, then with time-dependence and delayed neutrons.

The MNP method requires repeated evaluation of the full-order TH operator during inner iterations, which the NILO-CMFD method does not. If time allows, I would like to implement the MNP method using the same approximate TH operator in inner iterations.

I also plan to perform Fourier stability analysis to determine expected rate of convergence of the MNP method.

1.1 Problem Statement

The multigroup neutron transport equation in one spatial dimension with isotropic scattering is

$$\begin{aligned}
 & \left[\frac{1}{v_g} \frac{\partial}{\partial t} + \mu \frac{\partial}{\partial z} + \Sigma_t(z) \right] \psi_g(z, \mu, t) \\
 &= \frac{\chi_g^p(z, t)}{2} \sum_{g'=1}^{N_g} (1 - \beta_{g'}) \nu_g(z) \Sigma_{f,g'}(z) \phi_{g'}(z, t) + \frac{1}{2} \sum_{g'=1}^{N_g} \Sigma_{s,g' \rightarrow g}(z) \phi_{g'}(z, t) \\
 & \quad + \frac{1}{2} \sum_{k=1}^6 \lambda_k \chi_{g,k}^d(z, t) C_k^d(z, t) + q_g(z, \mu, t), \\
 & \quad g = 1, \dots, N_g, \quad 0 \leq z \leq Z, \quad -1 \leq \mu \leq 1, \quad t > 0 \quad (1a)
 \end{aligned}$$

$$\psi_g(z, \mu, 0) = \psi^0, \quad (1b)$$

$$\psi_g(0, \mu, t) = \psi_{in,g,0}, \quad \mu > 0 \quad \psi_g(Z, \mu, t) = \psi_{in,g,Z}, \quad \mu < 0, \quad (1c)$$

where the scalar flux is

$$\phi_g(z, t) = \int_{-1}^1 d\mu \psi_g(z, \mu, t). \quad (2)$$

and the cross-sections are temperature- and power-dependent,

$$\Sigma_u(z) = \Sigma_u(T(z, t), h(z, t)), \quad (3)$$

and

- z = spatial position
- μ = direction cosine
- t = time
- v_g = particle velocity in group g
- ψ_g = angular flux in group g
- ϕ_g = scalar flux in group g
- $\Sigma_{u,g}$ = cross section of reaction u
- ν_g = fission multiplicity in group g
- χ_g^p = prompt neutron energy emission spectrum
- $\chi_{g,k}^d$ = delayed neutron energy emission spectrum from precursor group k
- β_g = delayed neutron fraction in group g

- C_k^d = delayed neutron precursor population, group k
- λ_k = group-averaged precursor decay constant, group k
- T = temperature
- h = fission reaction rate $\sum_g \Sigma_{f,g} \phi_g$

The delayed neutron precursors are governed by the equations

$$\frac{\partial}{\partial t} C_k^d(z, t) = -\lambda_k C_k^d(z, t) + \sum_{g=1}^{N_g} \beta_{k,g}(z, t) \nu_g \Sigma_{f,g}(z) \phi_g(z, t), \quad k = 1, \dots, 6 \quad (4a)$$

$$C_k^d(z, 0) = C_k^{d,0}, \quad k = 1, \dots, 6 \quad (4b)$$

where $\beta_{k,g}$ is the delayed neutron fraction of precursor group k in energy group g .

Heat transfer is governed by the general equation

$$\mathcal{H}T(z, t) = W \sum_{g=1}^{N_g} \Sigma_{f,g}(z) \phi_g(z, t), \quad (5a)$$

$$T(z, 0) = T^0, \quad (5b)$$

where

W = energy released per fission

Equations (1), (3), (4) and (5) form a system of equations which are nonlinearly coupled and are characterized by multiple time and length scales. The high dimensionality of Equation (1) largely determines the complexity of the problem.

In operator form, the transport equation is

$$\frac{\partial \psi}{\partial t} + \mathcal{L}[T] \psi = \mathcal{S}[T] \psi \quad (6)$$

and the TH equation is

$$\frac{\partial T}{\partial t} + \mathcal{H}T = \mathcal{Q}[T, \psi] \quad (7)$$

where $\mathcal{L}$ represents streaming and collision, $\mathcal{S}$ represents source from scatter, fission, and precursor decay, and $\mathcal{Q}$ represents fission power hW .

Backward euler time discretization produces an effective steady-state equation at each time step, with augmented cross section and source. The effective steady-state equations for time step j can be written

$$\tilde{\mathcal{L}}^j[T^j] \psi^j = \tilde{\mathcal{S}}^j[T^j] \psi^j \quad (8)$$

$$\tilde{\mathcal{H}}^j T^j = \tilde{\mathcal{Q}}^j[T^j, \boldsymbol{\psi}^j] \quad (9)$$

where

$$\boldsymbol{\psi} = [\psi_1, \dots, \psi_g, \dots, \psi_G]^\top \quad (10)$$

A standard approach to solving a coupled problem like this one would be Fixed-Point Iteration (FPI), in which an estimate $T^{(n)}$ is used to calculate $\psi^{(n+1)}$ using the transport equation. $\psi^{(n+1)}$ is then used to define the fission power distribution for a TH calculation which determines $T^{(n+1)}$. This procedure is repeated until convergence. FPI can converge very slowly in the case of tight coupling, and requires both a high-order transport calculation and a potentially costly TH calculation at each iteration.

The problem may also be solved using Newton's method or Jacobian-free Newton-Krylov methods, which still require solution of the high-order transport equation at each iteration. The solution of the transport equation may be aided with a synthetic acceleration method, and the coupling may be aided using a low-order approximation of the TH operator. This is the basis of the NILO-CMFD method, which converges at a comparable rate to single-physics problems, and introduces no approximation to the final solution.

The MNP method [1] uses the Quasidiffusion method to project the transport equation to the GLOQD equation, which depends only on space and time. The coupled system of the MLOQD equations, precursor balance, and TH is then solved with Newton iteration.

1.2 TH equations

The TH operator $\mathcal{H}$ can be treated using any TH code, but for this application $\mathcal{H}$ will be the time-dependent advection-diffusion operator,

$$\mathcal{H}T(z) = \left[\frac{\partial}{\partial t} - \frac{\partial}{\partial z} k_b \frac{\partial}{\partial z} + \dot{m} c_p \frac{\partial}{\partial z} \right] T(z), \quad (11)$$

where

- c_p = specific heat capacity
- k_b = turbulent diffusion coefficient
- $\dot{m}$ = mass flow rate

Cross section temperature dependence can be general (or even performed using table lookup) but for this application the temperature dependence of cross sections will be the model used in [2],

$$\Sigma_u(z) = \Sigma_{u,0} + \Sigma_{u,1,d} \left(\sqrt{T(z) + \kappa W h(z)} - \sqrt{T_{d,0}} \right) + \Sigma_{u,1,b} (T(z) - T_0) \quad (12)$$

where $\kappa, \Sigma_{u,0}, \Sigma_{u,1,d}, \Sigma_{u,1,b}, T_{d,0}$, and T_0 are material constants, defined for each neutron energy group g .

1.3 Indices and Notation

The following notation will be adopted for this project:

- j = time index
- m = angular index for method of discrete ordinates
- k = delayed neutron precursor group
- g = neutron energy group
- n, s, l, r = iterative index: outermost, $\dots$, innermost
- i = spatial cell index
- u = reaction index for cross sections

2 NILO-CMFD Method

The NILO-CMFD method is described in [2] and [3], which describes five methods, FPI-CMFD, R-CMFD, X-CMFD, and two versions of NILO-CMFD.

Each outer (n) iteration of the CMFD-based methods consists of:

1. High-Order transport sweep using $\Sigma_u(T^{(n)})$
2. Calculation of corrected diffusion coefficient
3. Low-order diffusion solve for $\phi^{(n+1)}$
4. TH solution for $T^{(n+1)}$

The methods differ in how the low-order diffusion equation is solved:

- Fixed-point iteration solves the low-order equation using explicit cross sections (the same cross sections that were used for the high-order equation). FPI can be unstable.
- R-CMFD is similar to FPI-CMFD, but uses a “relaxed” heat source in the TH calculation, where $h = \Sigma_f(\alpha\phi^{(n+1)} + (1 - \alpha)\phi^{(n)})$. R-CMFD converges more slowly than FPI and may still be unstable for optically thick media.
- X-CMFD uses implicitly evaluated cross sections, requiring another iterative level, or Newton’s method, to solve the coupled system of low-order diffusion and TH. The method is stable, and outer iterations converge at a rate comparable to single-physics transport, but the method requires multiple inversions of the TH operator $\mathcal{H}$ within each outer iteration.
- NILO-CMFD uses an approximation of temperature in the X-CMFD’s coupled system of equations: cross section temperature dependence is linearized about the cross section values from the beginning of the iteration, and an approximation to the TH operator $\mathcal{H}$ is used. The method converges at nearly the same rate as X-CMFD at a reduced computational cost, and introduces no approximation in the result.
 - The method can also be applied without approximating the cross section evaluation. This method is called NILO-CMFD-A in [2].

First, Equation (1) is solved for ψ , which is used to determine the corrected diffusion coefficients $\hat{D}_g$, and another cross section evaluation is performed. The coupled system of low-order corrected diffusion and TH is then solved,

$$-\frac{d}{dz}D^{(n)}(z)\frac{d\phi^{(n+1)}}{dz} + \left(\tilde{\Sigma}_a^{(n+1)}(z) + \hat{D}^{(n+1)}(z)\right)\phi^{(n+1)}(z) = \nu\tilde{\Sigma}_f^{(n+1)}(z)\phi^{(n+1)}(z), \quad (13a)$$

$$\mathcal{H}T^{(n+1)}(z) = W\tilde{\Sigma}_f^{(n+1)}\phi^{(n+1)}(z) \quad (13b)$$

Where the cross section dependence on temperature and flux are approximated by a first-order Taylor expansion,

$$\Sigma_u^{(n+1)} = \Sigma_u(T^{(n+1)}, h^{(n+1)}) \quad (14)$$

$$= \Sigma_u(T^{(n)} + (T^{(n+1)} - T^{(n)}), h^{(n)} + (h^{(n+1)} - h^{(n)})) \quad (15)$$

$$\approx T^{(n)} + \frac{\partial \Sigma_u^{(n)}}{\partial T}(T^{(n+1)} - T^{(n)}) + \frac{\partial \Sigma_u^{(n)}}{\partial h}(h^{(n+1)} - h^{(n)}), \quad (16)$$

and $T^{(n+1)} - T^{(n)}$ is approximated using an approximate operator $H \approx \mathcal{H}$:

$$\Sigma_u^{(n+1)} \approx \Sigma_u^{(n)} + \frac{\partial \Sigma_u^{(n)}}{\partial T} \mathcal{H}^{-1}W(\Sigma_f^{(n+1)}\phi^{(n+1)} - \Sigma_f^{(n)}\phi^{(n)}) \quad (17)$$

$$\begin{aligned} & + \frac{\partial \Sigma_u^{(n)}}{\partial h}(\Sigma_f^{(n+1)}\phi^{(n+1)} - \Sigma_f^{(n)}\phi^{(n)}) \\ \approx & \Sigma_u^{(n)} + \frac{\partial \Sigma_u^{(n)}}{\partial T} H^{-1}W(\Sigma_f^{(n+1)}\phi^{(n+1)} - \Sigma_f^{(n)}\phi^{(n)}) \\ & + \frac{\partial \Sigma_u^{(n)}}{\partial h}(\Sigma_f^{(n+1)}\phi^{(n+1)} - \Sigma_f^{(n)}\phi^{(n)}). \end{aligned} \quad (18)$$

As the outer iterations converge, the term $(\Sigma_f^{(n+1)}\phi^{(n+1)} - \Sigma_f^{(n)}\phi^{(n)})$ goes to zero and thus there is no approximation in the final result. The approximate operator H should be chosen such that that it is accurate when T has weak spatial variation. [3] proposes H to be the advection operator without diffusion.

This method requires that

- $\Sigma_f^{(n)}$ and $\phi^{(n)}$ must be stored from the previous outer iteration
- One additional cross section evaluation is performed for numerical differentiation

The cross section derivatives can be approximated using

$$\dot{\Sigma}_{u,T}^{(n)} = \left. \frac{\partial \Sigma_u}{\partial T} \right|_{T=T^{(n)}} \approx \frac{\Sigma_u(T^{(n)}(1 + \epsilon_T)) - \Sigma_u(T^{(n)})}{\epsilon_T T^{(n)}} \quad (19)$$

$$\dot{\Sigma}_{u,f}^{(n)} = \left. \frac{\partial \Sigma_u}{\partial f} \right|_{f=f^{(n)}} \approx \frac{\Sigma_u(f^{(n)}(1 + \epsilon_f)) - \Sigma_u(f^{(n)})}{\epsilon_f f^{(n)}}. \quad (20)$$

Thus, the cross-section approximation for the coupled low-order system in the NILO-CMFD method is

$$\begin{aligned}\tilde{\Sigma}_u^{(n+1)} = T^{(n)} &+ \dot{\Sigma}_{u,T}^{(n)} H^{-1} W (\Sigma_f^{(n+1)} \phi^{(n+1)} - \Sigma_f^{(n)} \phi^{(n)}) \\ &+ \dot{\Sigma}_{u,f}^{(n)} (\Sigma_f^{(n+1)} \phi^{(n+1)} - \Sigma_f^{(n)} \phi^{(n)}).\end{aligned}\quad (21)$$

After the coupled system has been solved, the full-order TH equation is solved to determine $T^{(n+1)}$.

The NILO-CMFD algorithm for a steady-state problem (or a single time step of a time-dependent problem) is presented in Algorithm 1.

Algorithm 1 NILO-CMFD

Input: Initial iterates $T^{(0)}$ and $\phi^{(0)}$

$\Sigma_u^{(0)} \leftarrow \Sigma_u(T^{(0)})$

$n \leftarrow 0$

while $\|T^{(n+1)} - T^{(n)}\|_\infty / \|T^{(n)}\|_\infty > \epsilon_n$ **do**

Outer iterations;

Calculate $\phi^{(n+1/2)}$

▷ High-order transport sweep

TH calculation for $T^{(n+1/2)}$

Cross section evaluation for $\Sigma_u^{(n+1/2)}$

Cross section evaluation for $\dot{\Sigma}_u^{(n+1/2)}$

▷ Calculated numerically

$s \leftarrow 0$

while $\|T^{(s+1)} - T^{(s)}\|_\infty / \|T^{(s)}\|_\infty > \epsilon_s$ **do**

$s \leftarrow s + 1$

Calculate $\phi^{(n+1,s+1)}$ using $\tilde{\Sigma}_u^{(n+1,s)}$

▷ Low-order (Corrected Diffusion)

Calculate approximate temperature update from $\tilde{\Sigma}^{(n+1,s)} \phi^{(n+1,s+1)} - \tilde{\Sigma}^{(n)} \phi^{(n)}$

Update cross sections $\tilde{\Sigma}^{(n+1,s+1)}$ from Taylor expansion

end while

$\phi^{(n+1)} \leftarrow \phi^{(n+1,s+1)}$

Update temperatures using full-order HT operator

end while

3 MNP Method

The Multilevel Nonlinear Projective method is formulated by projecting the full-order transport equation first onto the first two angular moment equations with exact closures, then onto an effective grey equation for energy-integrated scalar flux and current. The method introduces no approximation.

The first two angular moments ($\int \cdot d\mu$ and $\int \mu \cdot d\mu$) of the high-order transport equation are

$$\frac{1}{v_g} \frac{\partial \phi_g}{\partial t} + \frac{\partial J_g}{\partial z} + (\Sigma_{t,g} - \Sigma_{s,g \rightarrow g'}) \phi_g = \sum_{g'=1}^G \Sigma_{s,g' \rightarrow g} \phi_{g'} + \chi_g^p \overline{(1 - \beta) \nu \Sigma_f \phi} + q_{del,g}^x \quad (22a)$$

$$\frac{1}{v_g} \frac{\partial J_g}{\partial t} + \frac{\partial}{\partial z} (E_g \phi_g) + \Sigma_{t,g} J_g = 0, \quad (22b)$$

where, with vacuum boundaries,

$$J_g(0, t) = J_g(Z, t) = 0 \quad (22c)$$

$$\phi_g(z, 0) = \phi_g^0(z), \quad J_g(z, 0) = J_g^0(z) \quad (22d)$$

where E_g is the group Eddington factor,

$$E_g = \frac{\int_{-1}^1 \mu^2 \psi_g d\mu}{\int_{-1}^1 \psi_g d\mu}. \quad (23)$$

The Grey Low-Order Quasidiffusion (GLOQD) equations are:

$$\frac{\partial}{\partial t} \left(\frac{\phi}{\bar{v}} \right) + \frac{\partial J}{\partial z} + \left(\bar{\Sigma}_a - \overline{(1 - \beta)\nu\Sigma_f} \right) \phi = q_{del}^x \quad (24a)$$

$$\frac{\partial}{\partial t} \left(\frac{J}{\bar{v}} \right) + \frac{\partial}{\partial z} (\bar{E}\phi) + \bar{\Sigma}_{tr} J + \bar{\zeta}\phi = 0, \quad (24b)$$

where the initial and boundary conditions, as well as spectrum-averaged cross sections and factors $\bar{\Sigma}_a, \bar{E}, \bar{\zeta}, \bar{v}$ are defined in [1].

In each outer iteration of the MNP method, the following is performed:

1. Cross-sections are evaluated from previous iteration's temperature $T^{(n)}$
2. High-order transport sweep (eq. (1)) is used to determine ψ_g , which is used to calculate group Eddington factor
3. Multigroup low-order Quasidiffusion equations (eq. (22)) are solved for ϕ_g, J_g , which are used to calculate spectrum-averaged cross sections, Eddington factor, and velocity
4. Solve the coupled GLOQD (eq. (24)) and TH equations to yield $\phi^{(n+1)}$ and $T^{(n+1)}$

A sketch of the Multilevel Nonlinear Projective method is presented in Algorithm 2. The full algorithm is presented in [1], which also gives the Multigroup Low-Order Quasidiffusion (MLOQD) and Grey LOQD (GLOQD) equations which are necessary for its implementation.

4 Preliminary and expected results

My implementation of the MNP method will use the Linear Discontinuous (LD) finite element discretization in 1D slab geometry (derived in Section A), and finite volume discretization of the TH operator (for which I will use the advection-diffusion equation, as in [3]).

I will use Jacobian-free Newton-Krylov iteration for the innermost iterations of Algorithm 2 for MNP implementation.

Algorithm 2 MNP

Input: Initial iterates $T^{(0)}$ and $\phi^{(0)}$

$n \leftarrow 0$

while $\|T^{(n+1)} - T^{(n)}\|_\infty / \|T^{(n)}\|_\infty > \epsilon_1$ **do**

Outer Iterations;

$n \leftarrow n + 1$

Calculate multigroup cross sections

Calculate effective grey cross sections

Calculate $\psi^{(n+1)}$ using $\phi^{(n)}$; iterate through groups only once $\triangleright$ High-order transport

Calculate $E_g^{(n+1)}, B_g^{(n+1)}$

$s \leftarrow 0$

while $\|T^{(n,s+1)} - T^{(n,s)}\|_\infty / \|T^{(n,s)}\|_\infty > \epsilon_2$ **do**

Multigroup QD iterations;

$s \leftarrow s + 1$

Update cross sections, fission power source

Solve MLOQD equations for $\phi^{(n+1,s+1)}$ and $J^{(n+1,s+1)}$

Calculate effective grey closure terms and cross sections

$l \leftarrow 0$

while $\|T^{(n,s+1,l+1)} - T^{(n,s+1,l)}\|_\infty / \|T^{(n,s+1,l)}\|_\infty > \epsilon_3$ **do**

Newton iterations;

$l \leftarrow l + 1$

Solve the system GLOQD, $\Sigma(T)$, and linearized full-order TH.

end while

end while

end while

A time-dependent transient benchmark is introduced in [1], which I can compare my results to. If I implement both the MNP and NILO-MNP methods, I will record the number of each kind of iteration required (i.e. to count number of $n \times n$ linear solves required for each method) and residual histories for each method to show convergence rate.

If time allows for a Fourier stability analysis, the intended results are:

- Region of stability for the NILO-MNP method
- Spectral radii for FPI-MNP, MNP, and NILO-MNP methods

References

- [1] A. Tamang and D. Y. Anistratov, “A multilevel projective method for solving the space-time multigroup neutron kinetics equations coupled with the heat transfer equation,” *Nuclear Science and Engineering*, vol. 177, no. 1, pp. 1–18, 2014.
- [2] N. J. Adamowicz, A. Manera, and E. W. Larsen, “The NILO-CMFD method for iteratively solving coupled neutron transport–thermal hydraulics problems,” *Nuclear Science and Engineering*, vol. 197, no. 2, pp. 262–278, 2023.
- [3] “The NILO-CMFD method for iteratively solving multiphysics neutron transport - thermal hydraulic problems,” in *Proceedings of the American Nuclear Society Conference on Mathematics and Computation 2021*, pp. 2092–2101, ANS, 2021.

A LD spatial discretization of the transport equation

Using some shorthand notation, the multigroup 1D diffusion equation, after applying the Method of Discrete Ordinates and backward Euler temporal differencing, is, for angle m ,

$$\mu_m \frac{\partial}{\partial z} \psi_{m,g} + \hat{\Sigma}_{t,g} \psi_{m,g} = \frac{\chi_g}{2} \sum_{g'=1}^{N_g} c_{f,g'} \phi_{g'} + \frac{1}{2} \sum_{g'=1}^{N_g} \sigma_{s,g' \rightarrow g} \phi_{g'} + \hat{q}_{m,g} \quad (25)$$

where $\hat{\Sigma}_{t,g}$ includes contribution from the temporal discretization, and $\hat{q}$ includes contribution from temporal discretization and the precursor decay source. The coefficient c_f is

$$c_{f,g} = (1 - \beta_g) \nu \Sigma_{f,g}. \quad (26)$$

The spatial grid $\{\kappa_i^z\}_{i=1}^{N_i}$ is introduced, where

$$\kappa_i^z = [z_{i-1/2}, z_{i+1/2}] \quad (27)$$

$$z_{1/2} = 0 \quad (28)$$

$$z_{N_i+1/2} = Z. \quad (29)$$

Then, approximating each spatial function as a combination of linear basis functions,

$$\psi_{m,g}(z) = \sum_{c=L,R} \psi_{m,g,c,i} b_{c,i}, \quad z \in \kappa_i^z \quad (30)$$

$$b_{L,i} = \frac{z_{i+1/2} - z}{\Delta z_i} \quad (31)$$

$$b_{R,i} = \frac{z - z_{i-1/2}}{\Delta z_i}, \quad (32)$$

where

$$\int_{\kappa_i} b_{L,i}^2 dz = \int_{\kappa_i} b_{R,i}^2 dz = \frac{1}{3} \Delta z_i \quad (33)$$

$$\int_{\kappa_i} b_{L,i} dz = \int_{\kappa_i} b_{R,i} dz = \frac{1}{2} \Delta z_i \quad (34)$$

$$\int_{\kappa_i} b_{L,i} b_{R,i} dz = \frac{1}{6} \Delta z_i \quad (35)$$

Then, eq. (25) is integrated over the two basis functions in each cell. Note that this requires the cross sections (and thus temperature) to be constant in each spatial cell.

$$\begin{aligned} \mu_m \int_{\kappa_i} \frac{\partial}{\partial z} \psi_{m,g} b_{L,i} dz + \hat{\Sigma}_{t,g} \int_{\kappa_i} \psi_{m,g} b_{L,i} dz = \\ \frac{\chi_g}{2} \sum_{g'=1}^{N_g} c_{f,g'} \int_{\kappa_i} \phi_{g'} b_{L,i} dz + \frac{1}{2} \sum_{g'=1}^{N_g} \sigma_{s,g' \rightarrow g} \int_{\kappa_i} \phi_{g'} b_{L,i} dz + \int_{\kappa_i} \hat{q}_{m,g} b_{L,i} dz \end{aligned} \quad (36a)$$

$$\begin{aligned} \mu_m \int_{\kappa_i} \frac{\partial}{\partial z} \psi_{m,g} b_{R,i} dz + \hat{\Sigma}_{t,g} \int_{\kappa_i} \psi_{m,g} b_{R,i} dz = \\ \frac{\chi_g}{2} \sum_{g'=1}^{N_g} c_{f,g'} \int_{\kappa_i} \phi_{g'} b_{R,i} dz + \frac{1}{2} \sum_{g'=1}^{N_g} \sigma_{s,g' \rightarrow g} \int_{\kappa_i} \phi_{g'} b_{R,i} dz + \int_{\kappa_i} \hat{q}_{m,g} b_{R,i} dz \end{aligned} \quad (36b)$$

Then the divergence theorem is applied to the first term on the right-hand side,

$$\begin{aligned} \mu_m [\psi_{m,g}(z) b_{L,i}]_{z_{i-1/2}}^{z_{i+1/2}} - \mu_m \int_{\kappa_i} \psi_{m,g} \frac{\partial}{\partial z} b_{L,i} dz + \hat{\Sigma}_{t,g} \int_{\kappa_i} \psi_{m,g} b_{L,i} dz = \\ \frac{\chi_g}{2} \sum_{g'=1}^{N_g} c_{f,g'} \int_{\kappa_i} \phi_{g'} b_{L,i} dz + \frac{1}{2} \sum_{g'=1}^{N_g} \sigma_{s,g' \rightarrow g} \int_{\kappa_i} \phi_{g'} b_{L,i} dz + \int_{\kappa_i} \hat{q}_{m,g} b_{L,i} dz \end{aligned} \quad (37a)$$

$$\begin{aligned} \mu_m [\psi(z)_{m,g} b_{R,i}]_{z_{i-1/2}}^{z_{i+1/2}} - \mu_m \int_{\kappa_i} \psi_{m,g} \frac{\partial}{\partial z} b_{R,i} dz + \hat{\Sigma}_{t,g} \int_{\kappa_i} \psi_{m,g} b_{R,i} dz = \\ \frac{\chi_g}{2} \sum_{g'=1}^{N_g} c_{f,g'} \int_{\kappa_i} \phi_{g'} b_{R,i} dz + \frac{1}{2} \sum_{g'=1}^{N_g} \sigma_{s,g' \rightarrow g} \int_{\kappa_i} \phi_{g'} b_{R,i} dz + \int_{\kappa_i} \hat{q}_{m,g} b_{R,i} dz \end{aligned} \quad (37b)$$

For the boundary values, $\psi(x_{i\pm 1/2})$, the upwind flux is defined as

$$\psi_{m,g,L,i}^b = \begin{cases} \psi_{m,g,R,i-1}, & \mu > 0 \\ \psi_{m,g,L,i}, & \mu < 0 \end{cases}, \quad \psi_{m,g,R,i}^b = \begin{cases} \psi_{m,g,R,i}, & \mu > 0 \\ \psi_{m,g,L,i+1}, & \mu < 0 \end{cases} \quad (38)$$

then, using the upwinding scheme and expanding each function in terms of its basis functions,

$$\begin{aligned} \mu_m [\psi_{m,g,R,i}^b - \psi_{m,g,L,i}^b] + \frac{1}{2} \mu_m (\psi_{m,g,L,i} + \psi_{m,g,R,i}) + \hat{\Sigma}_{t,g} \frac{1}{6} \Delta z_i (2\psi_{m,g,L,i} + \psi_{m,g,R,i}) = \\ \frac{\chi_g}{2} \sum_{g'=1}^{N_g} c_{f,g'} \frac{1}{6} \Delta z_i (2\phi_{g',L,i} + \phi_{g',R,i}) + \frac{1}{2} \sum_{g'=1}^{N_g} \sigma_{s,g' \rightarrow g} \frac{1}{6} \Delta z_i (2\phi_{g',L,i} + \phi_{g',R,i}) \\ + \frac{1}{6} \Delta z_i (2q_{m,g,L,i} + q_{m,g,R,i}) \end{aligned} \quad (39a)$$

$$\begin{aligned}
\mu_m [\psi_{m,g,R,i}^b - \psi_{m,g,L,i}^b] - \frac{1}{2}\mu_m (\psi_{m,g,L,i} + \psi_{m,g,R,i}) + \hat{\Sigma}_{t,g} \frac{1}{6} \Delta z_i (\psi_{m,g,L,i} + 2\psi_{m,g,R,i}) = \\
\frac{\chi_g}{2} \sum_{g'=1}^{N_g} c_{f,g'} \frac{1}{6} \Delta z_i (\phi_{g',L,i} + 2\phi_{g',R,i}) + \frac{1}{2} \sum_{g'=1}^{N_g} \sigma_{s,g' \rightarrow g} \frac{1}{6} \Delta z_i (\phi_{g',L,i} + 2\phi_{g',R,i}) \\
+ \frac{1}{6} \Delta z_i (q_{m,g,L,i} + 2q_{m,g,R,i}) \quad (39b)
\end{aligned}$$